MEDITERRANEAN COLLEGE · 6CM521

Frontiers in AI and Data Science

Interactive course materials: lectures, labs, notebooks and datasets. Everything runs in your web browser. There is nothing to install.

OPEN THE COURSE MATERIALS

invariantum.github.io/frontiers_in_ai_and_data_science/lab/index.html

Click the link, or scan the code with your phone. Bookmark it: this is the only link you need for the whole module.

Getting started (first time, about 2 minutes)

1. **Use a laptop or desktop with Chrome, Edge or Firefox**
   Open the link in a **normal** browser window, **not** a private or incognito window (the notebooks cannot see their data files there). Phones and tablets work, but a computer is much easier.
2. **Wait for the first load**
   The first time, it takes **10–30 seconds** while Python downloads into your browser. After that it opens in a few seconds.
3. **Check that everything works**
   In the file list on the left, open **00_start_here** and double-click **00_environment_check.ipynb**, then choose **Run** → **Run All Cells**. Every line should say **OK**. If anything says FAILED, take a screenshot and send it to your lecturer.
4. **Open a lab notebook**
   Go into the folder for the week (for example **week02_data_preprocessing** → **lab**) and double-click a notebook (**.ipynb**); they are numbered in the order to work through them. Run one cell with **Shift + Enter**, or all cells with **Run** → **Run All Cells**. The datasets are already in the same folder, so they load automatically.

Saving your work

Your changes are saved **inside this browser, on this computer only**. They do not appear on another computer or browser, and they are deleted if you clear your browser data.

To keep your work: right-click the notebook in the file list → **Download**. Do this at the end of every session, and always before submitting work.

Notebooks to read, not run

Four advanced notebooks need a GPU or large AI libraries that cannot run in a browser. Open them to read the code and the saved results; your lecturer will demonstrate them in class. Each one says **Read-only in the browser** at the top.

Notebook	Folder
QLoRA_customer_support	extras
YOLOv8_object_detection_on_custom_dataset	extras
object-detection-demo-in-tensorflow-and-yolo-v3	week15-16_object_detection_explainability → lab
week-4-bd-anchor-shap-lime-eli5-4-rf-explaining	week15-16_object_detection_explainability → lab

What's inside

Folder	Topics
00_start_here	Environment check, assessment brief, useful links
week01_introduction	Introduction to the module, Python crash course and exercises
week02_data_preprocessing	Data preprocessing with Pandas: Series, DataFrames, missing data, groupby, merging, input/output
week03_exploratory_data_analysis	Exploratory data analysis (e-commerce and Titanic examples)
week04_frequent_patterns	Frequent patterns, association rules, correlation analysis
week05_regression	Regression; Weeks 5–8 lab sheet (regression)
week06_classification_decision_trees	Classification and decision trees; Weeks 5–8 lab sheet
week07_random_forests_knn	Random forests and KNN; Weeks 5–8 lab sheet (KNN)
week08_clustering	Clustering (unsupervised learning); Weeks 5–8 lab sheet
week09_generative_ai_prompting	Generative AI and prompt engineering; LLM APIs lab
week10_reinforcement_learning	Reinforcement learning
week11_reinforcement_learning_lab	Reinforcement learning lab (FrozenLake)
week13-14_neural_networks_xai	Neural networks and explainable AI; fine-tuning and evaluation lab
week15-16_object_detection_explainability	Object detection; explaining models with SHAP, LIME, ELI5, Anchor
extras	Case studies: RAG + MCP + LLMs, agentic AI for economics, LoRA / QLoRA, YOLOv8

Every week has the same layout: **lecture** (slides as PDF to view in the browser, plus PowerPoint) and **lab** (notebooks, their data and lab sheets). To save any file, right-click it → **Download**.

If something goes wrong

Problem	What to do
A notebook cannot find its data file (*FileNotFoundError*)	You are probably in a private/incognito window. Open the link in a normal window and reload.
A notebook looks out of date or different from the lecture	Your browser is showing your own older saved copy. Download it if you want to keep your changes, then right-click the notebook → **Delete** and reload the page. The current version reappears.
Nothing happens when you run a cell, or it seems frozen	Choose **Kernel** → **Restart Kernel**, then run the cells again from the top.
An error appears in a Week 1 exercise	Some cells show an error on purpose (for example, changing a tuple), and exercise cells stay empty until you write the code yourself.
The page stays on “Loading”	Wait 30 seconds, then reload. Check your internet connection.